Chetanya
1024030027
3C35

Question 1:

Write Java Program to Check Leap Year

Code:

```
public class Q1 {

   static boolean isLeap(int year) {
      if (year % 400 == 0) {
         return true;
      } else if (year % 100 == 0) {
         return false;
      } else if (year % 4 == 0) {
         return true;
      }
      return false;
   }

   public static void main(String[] args) {
      int[] years = {2024, 1900, 2000, 2026};
      for (int y : years) {
         if (isLeap(y)) {
            System.out.println(y + " is a Leap Year");
         } else {
            System.out.println(y + " is not a Leap Year");
         }
      }
   }
}
```

Output:

```
2024 is a Leap Year
1900 is not a Leap Year
2000 is a Leap Year
2026 is not a Leap Year
```

Question 2:

Write Java Program to Check Whether a Character is Alphabet or Not

Code:

```
public class Q2 {
   public static void main(String[] args) {
      char[] chars = {'a', 'Z', '7', '$'};
      for (char c : chars) {
```

```
            if ((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z')) {
                System.out.println(c + " is an Alphabet");
            } else {
                System.out.println(c + " is not an Alphabet");
            }
        }
    }
}
```

Output:

```
a is an Alphabet
Z is an Alphabet
7 is not an Alphabet
$ is not an Alphabet
```

Question 3:

Write Java Program to Find Factorial of a Number

Code:

```
public class Q3 {

    static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    public static void main(String[] args) {
        int[] nums = {0, 5, 10};
        for (int n : nums) {
            System.out.println("Factorial of " + n + " = " + factorial(n));
        }
    }
}
```

Output:

```
Factorial of 0 = 1
Factorial of 5 = 120
Factorial of 10 = 3628800
```

Question 4:

Write Java Program to Display Fibonacci Series

Code:

```
public class Q4 {
    public static void main(String[] args) {
        int n = 10;
```

```
        int a = 0, b = 1;
        System.out.print("Fibonacci Series up to " + n + " terms: ");
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
        System.out.println();
    }
}
```

Output:

Fibonacci Series up to 10 terms: 0 1 1 2 3 5 8 13 21 34

Question 5:

Write Java Program to Find GCD of two Numbers

Code:

```
public class Q5 {

    static int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

    public static void main(String[] args) {
        int a = 36, b = 60;
        System.out.println("GCD of " + a + " and " + b + " = " + gcd(a, b));
        a = 17;
        b = 5;
        System.out.println("GCD of " + a + " and " + b + " = " + gcd(a, b));
    }
}
```

Output:

GCD of 36 and 60 = 12
GCD of 17 and 5 = 1

Question 6:

Write Java Program to Find LCM of two Numbers

Code:

```
public class Q6 {
```

```
    static int gcd(int a, int b) {
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

    static int lcm(int a, int b) {
        return (a * b) / gcd(a, b);
    }

    public static void main(String[] args) {
        int a = 12, b = 18;
        System.out.println("LCM of " + a + " and " + b + " = " + lcm(a, b));
        a = 7;
        b = 9;
        System.out.println("LCM of " + a + " and " + b + " = " + lcm(a, b));
    }
}
```

Output:

LCM of 12 and 18 = 36
LCM of 7 and 9 = 63

Question 7:

Write Java Program to Count Number of Digits in an Integer

Code:

```
public class Q7 {

    static int countDigits(int n) {
        if (n == 0) {
            return 1;
        }
        n = Math.abs(n);
        int count = 0;
        while (n > 0) {
            n = n / 10;
            count++;
        }
        return count;
    }

    public static void main(String[] args) {
        int[] nums = {12345, 0, -9876, 7};
        for (int n : nums) {
```

```
            System.out.println("Number of digits in " + n + " = " + countDigits(n));
        }
    }
}
```

Output:

Number of digits in 12345 = 5
Number of digits in 0 = 1
Number of digits in -9876 = 4
Number of digits in 7 = 1

Question 8:

Write Java Program to Reverse a Number

Code:

```
public class Q8 {

    static int reverse(int n) {
        int rev = 0;
        while (n != 0) {
            int digit = n % 10;
            rev = rev * 10 + digit;
            n = n / 10;
        }
        return rev;
    }

    public static void main(String[] args) {
        int[] nums = {12345, 1200, 908};
        for (int n : nums) {
            System.out.println("Reverse of " + n + " = " + reverse(n));
        }
    }
}
```

Output:

Reverse of 12345 = 54321
Reverse of 1200 = 21
Reverse of 908 = 809

Question 9:

Write Java Program to Calculate the Power of a Number

Code:

```
public class Q9 {

    static long power(int base, int exp) {
        long result = 1;
        for (int i = 1; i <= exp; i++) {
```

```
            result = result * base;
        }
        return result;
    }

    public static void main(String[] args) {
        System.out.println("2 ^ 10 = " + power(2, 10));
        System.out.println("5 ^ 3 = " + power(5, 3));
        System.out.println("7 ^ 0 = " + power(7, 0));
        System.out.println("Using Math.pow: 3 ^ 4 = " + (int) Math.pow(3, 4));
    }
}
```

Output:

```
2 ^ 10 = 1024
5 ^ 3 = 125
7 ^ 0 = 1
Using Math.pow: 3 ^ 4 = 81
```

Question 10:

Write Java Program to Check Palindrome

Code:

```
public class Q10 {

    static boolean isPalindrome(int n) {
        int original = n;
        int rev = 0;
        while (n > 0) {
            rev = rev * 10 + n % 10;
            n = n / 10;
        }
        return original == rev;
    }

    static boolean isPalindrome(String s) {
        s = s.toLowerCase();
        int i = 0, j = s.length() - 1;
        while (i < j) {
            if (s.charAt(i) != s.charAt(j)) {
                return false;
            }
            i++;
            j--;
        }
        return true;
    }
```

```
    public static void main(String[] args) {
        int[] nums = {121, 12321, 123};
        for (int n : nums) {
            System.out.println(n + (isPalindrome(n) ? " is a Palindrome" : " is not a Palindrome"));
        }
        String[] words = {"Madam", "racecar", "java"};
        for (String w : words) {
            System.out.println(w + (isPalindrome(w) ? " is a Palindrome" : " is not a Palindrome"));
        }
    }
}
```

Output:

121 is a Palindrome
12321 is a Palindrome
123 is not a Palindrome
Madam is a Palindrome
racecar is a Palindrome
java is not a Palindrome